Week 4: Adding an Animation to an App

Animations will increase the user experience. Flutter has two types of animations. Physics-based and Tweens. Physics-based animation is used to mimic real-world behavior. For example, when an object is dropped and hits the ground, it will bounce and continue to move forward, but with each bounce, it continues to slow down with smaller rebounds and eventually stop. As the object gets closer to the ground with each bounce, the velocity increases, but the height of the bounce decreases.
Tween is short for "in-between," meaning that the animation has beginning and ending points, a timeline, and a curve that specifies the timing and speed of the transition. The beauty is that the framework automatically calculates the transition from the beginning to end point.

USING ANIMATEDCONTAINER

Let's start with a simple animation by using the `AnimatedContainer` widget. This is a `Container` widget that gradually changes values over a period of time. The `AnimatedContainer` constructor has arguments called `duration`, `curve`, `color`, `height`, `width`, `child`, `decoration`, `transform`, and many others.

1. Create a Flutter project.
2. Add the following code

```
import 'package:flutter/material.dart';

class AnimatedContainerWidget extends StatefulWidget {
    const AnimatedContainerWidget({
    Key key,
    }) : super(key: key);
    @override
    _AnimatedContainerWidgetState createState() => _AnimatedContainerWidgetState();
    }
    class _AnimatedContainerWidgetState extends State<AnimatedContainerWidget> {
    double _height = 100.0;
    double _width = 100.0;
    void _increaseWidth() {
    setState(() {
    _width = _width >= 320.0 ? 100.0 : _width += 50.0;
    });
    }

    @override
    Widget build(BuildContext context) {
    return Row(
    children: <Widget>[
    AnimatedContainer(
    duration: Duration(milliseconds: 500),
    curve: Curves.elasticOut,
    color: Colors.amber,
    height: _height,
    width: _width,
    child: FlatButton(
    child: Text('Tap to\nGrow Width\n$_width'),
    onPressed: () {
    _increaseWidth();
    },
    ),
    ),
    ],
    );
    }
    }
```

3. Open the `home.dart` file and import the `animated_container.dart` file to the top of the page.

```
import 'package:flutter/material.dart';
```

```
import 'package:ch7_animations/widgets/animated_container.dart';
```

4. Add to the body a `SafeArea` widget with `Column` as a `child`. In the `Column` children, add the call to the widget class `AnimatedContainerWidget()`.

```
body: SafeArea(
child: Column(
children: <Widget>[
AnimatedContainerWidget(),
],
),
),
```

USING ANIMATEDCROSSFADE

The `AnimatedCrossFade` widget provides a great cross-fade between two children widgets. The `AnimatedCrossFade` constructor takes `duration`, `firstChild`, `secondChild`, `crossFadeState`, `sizeCurve`, and many other arguments.

1. Create a new Dart file in the `widgets` folder. Right-click the `widgets` folder, select New ➪ Dart File, enter **`animated_cross_fade.dart`**, and click the OK button to save.

2. Import the `material.dart` library, add a new line, and start typing **`st`**. The autocompletion help opens. Select the `stful` abbreviation and give it a name of `AnimatedCrossFadeWidget`.

3. After the class `_AnimatedCrossFadeWidgetState extends State<AnimatedCrossFadeWidget>` and before `@override`, add a variable for `_crossFadeStateShowFirst` and the `_crossFade()` method.
   The `_crossFadeStateShowFirst` variable is of type Boolean (`bool`).
   `bool _crossFadeStateShowFirst = true;`
   The `_crossFade()` method calls `setState()` to notify the framework that the `_crossFadeState-ShowFirst` value has changed and schedules a build for the state of this object to redraw the subtree. If you do not call `setState()`, the `_width` value still changes, but the `AnimatedCrossFade` widget will not redraw with the new value.

```
class _AnimatedCrossFadeWidgetState extends State<AnimatedCrossFadeWidget> {
    bool _crossFadeStateShowFirst = true;

void _crossFade() {
setState(() {
_crossFadeStateShowFirst = _crossFadeStateShowFirst ? false : true;
});
}
@override
Widget build(BuildContext context) {
```

   When the page is loaded, the `_crossFadeStateShowFirst` variable is initiated with a value of `true`. When `FlatButton` is tapped, the `onPressed` property calls the `_crossFade()` method. To track which child (`firstChild`, `secondChild`) the `AnimatedCrossFade` widget should show and animate, use the ternary operator. If the `_crossFadeStateShowFirst` is `true`, then set the `_crossFadeStateShowFirst` value to `false`. Otherwise, set it to `true` since the current value is `false`.

4. To keep the UI clean, add each animation widget in a separate `Row`. Embed the

`AnimatedCrossFade` widget in a `Row` with a `Stack` as a child. The reason to use a `Stack` is to add a `FlatButton` over the `AnimatedCrossFade` widget to give it a label and an `onPressed` event. You could also use a `GestureDetector` widget.

```
@override
Widget build(BuildContext context) {
return Row(
children: <Widget>[
Stack(
alignment: Alignment.center,
children: <Widget>[
AnimatedCrossFade(
duration: Duration(milliseconds: 500),
sizeCurve: Curves.bounceOut,
crossFadeState: _crossFadeStateShowFirst ? CrossFadeState.showFirst :
CrossFadeState.showSecond,
firstChild: Container(
color: Colors.amber,
height: 100.0,
width: 100.0,
),
secondChild: Container(
color: Colors.lime,
height: 200.0,
width: 200.0,
),
),
Positioned.fill(
child: FlatButton(
child: Text('Tap to\nFade Color & Size'),
onPressed: () {
_crossFade();
},
),
),
          ],
),
],
);
}
```

5. Use a duration of 500 milliseconds like in the previous animation.

```
duration: Duration(milliseconds: 500),
```

6. For `sizeCurve`, use `Curves.bounceOut` to see how the `Curves` class affects animations.

```
sizeCurve: Curves.bounceOut,
```

7. To decide which `Container` widget to show when the animation is completed, set the `cross-FadeState` argument value by using the ternary operator to check whether the `_crossFadeState-ShowFirst` value is true; then show `CrossFadeState.showFirst`; otherwise, show `CrossFadeState.showSecond`.

```
crossFadeState: _crossFadeStateShowFirst ? CrossFadeState.showFirst :
CrossFadeState.showSecond,
```

8. The animation will cross-fade between colors and sizes; for `firstChild` and `secondChild`, use a `Container`. The `firstChild Container` has a `Colors.amber` color property and `height` and `width` values of 100.0 pixels. The `secondChild Container` has a `Colors.lime` color property and `height` and `width` values of 200.0 pixels.

```
firstChild: Container(
color: Colors.amber,
height: 100.0,
width: 100.0,
),
secondChild: Container(
color: Colors.lime,
height: 200.0,
width: 200.0,
),
```

Add the second `Stack` child and call the `Positioned.fill` constructor with a child of `FlatButton`. Using `Positioned.fill` allows the `FlatButton` widget to resize itself to the maximum size of the `Stack` widget. For the `onPressed` property, add a call to the `_cross-Fade()` method.

```
Positioned.fill(
child: FlatButton(
child: Text('Tap to\nFade Color & Size'),
onPressed: () {
_crossFade();
},
),
),
```

Here's the full `_AnimatedCrossFadeWidget()` widget class:

```
import 'package:flutter/material.dart';
class AnimatedCrossFadeWidget extends StatefulWidget {
const AnimatedCrossFadeWidget({

Key key,
}) : super(key: key);
@override
_AnimatedCrossFadeWidgetState createState() => _AnimatedCrossFadeWidgetState();
}
class _AnimatedCrossFadeWidgetState extends State<AnimatedCrossFadeWidget> {
bool _crossFadeStateShowFirst = true;
void _crossFade() {
setState(() {
_crossFadeStateShowFirst = _crossFadeStateShowFirst ? false : true;
});
}
@override
Widget build(BuildContext context) {
return Row(
children: <Widget>[
Stack(
alignment: Alignment.center,
children: <Widget>[
AnimatedCrossFade(
duration: Duration(milliseconds: 500),
sizeCurve: Curves.bounceOut,
crossFadeState: _crossFadeStateShowFirst ? CrossFadeState.showFirst :
CrossFadeState.showSecond,
firstChild: Container(
color: Colors.amber,
height: 100.0,
width: 100.0,
),
secondChild: Container(
color: Colors.lime,
height: 200.0,
width: 200.0,
),
),
Positioned.fill(
child: FlatButton(
child: Text('Tap to\nFade Color & Size'),
onPressed: () {
_crossFade();
},
),
),
],
),
],
);
}
}
```

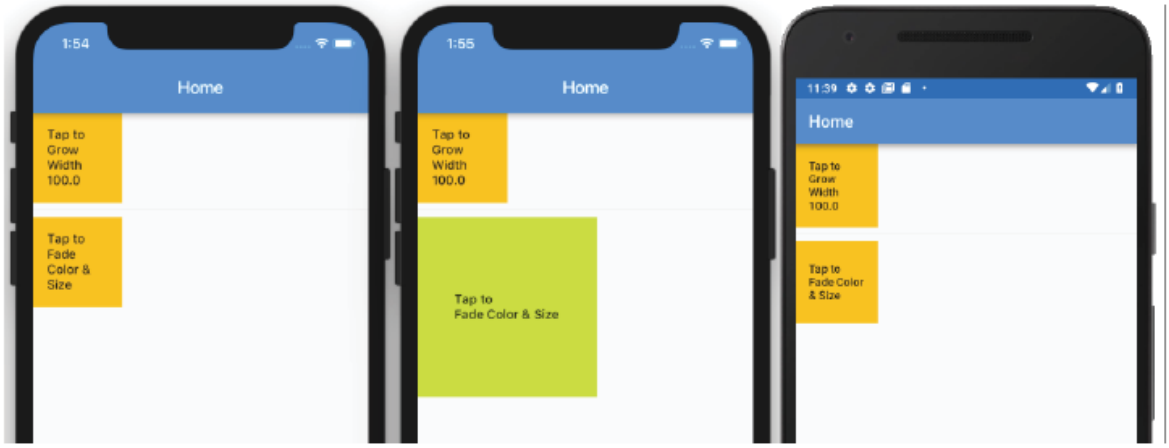
10. Continue editing the `home.dart` file and import the `animated_cross_fade.dart` file to the top of the page.

```
import 'package:flutter/material.dart';
```

```
import 'package:ch7_animations/widgets/animated_container.dart';
import 'package:ch7_animations/widgets/animated_cross_fade.dart';
```

11. Just after the `AnimatedContainerWidget()` widget class, add a `Divider()` widget. On the next line, add a call to the widget class `AnimatedCrossFadeWidget()`.

```
body: SafeArea(
child: Column(
children: <Widget>[
AnimatedContainerWidget(),
Divider(),
AnimatedCrossFadeWidget(),
],
),
),
```



USING ANIMATEDOPACITY

If you need to hide or partially hide a widget, `AnimatedOpacity` is a great way to animate fading over time. The `AnimatedOpacity` constructor the takes `duration`, `opacity`, `curve`, and `child` arguments. For this example, you do not use a curve; since you want a smooth fade-out and fade-in, it's not necessary.

Adding the AnimatedOpacity Widget

The example uses `opacity` to partially fade out a `Container` widget. The widget will animate the `opacity` value from fully visible to almost faded out.

1. Create a new Dart file in the `widgets` folder. Right-click the `widgets` folder, select New ➪ Dart File, enter **`animated_opacity.dart`**, and click the OK button to save.

2. Import the `material.dart` library, add a new line, and then start typing **`st`**. The autocompletion help opens. Select the `stful` abbreviation and give it a name of `AnimatedOpacityWidget`.

3. After the class `_AnimatedOpacityWidgetState extends State<AnimatedOpacityWidget>` and before `@override`, add the variable for `_opacity` and the `_animatedOpacity()` method. The `_opacity` variable is of type `double`.

```
double _opacity = 1.0;
```

The `_animatedOpacity()` method calls `setState()` to notify the framework that the `_opacity` value has changed and schedules a build for the state of this object to redraw the subtree. If you do not call `setState()`, the `_opacity` value still changes, but the `AnimatedOpacity` widget does not redraw with the new value.

```
class _AnimatedOpacityWidgetState extends State<AnimatedOpacityWidget> {
double _opacity = 1.0;
void _animatedOpacity() {
setState(() {
_opacity = _opacity == 1.0 ? 0.3 : 1.0;
});
}
@override
Widget build(BuildContext context) {
```

When the page is loaded, the `_opacity` variable is initiated with a value of 1.0, which is fully visible. When the `FlatButton` widget is tapped, the `onPressed` property calls the `_animatedOpacity()`

method.
To calculate the widget opacity, use the ternary operator. If the `_opacity` value is `1.0`, then set `_opacity` to `0.3`. Otherwise, set it to `1.0` since the current value is `0.3`.

To keep the UI clean, add each animation widget in a separate `Row`. Embed the `AnimatedOpacity` widget in a `Row`.

```
@override
Widget build(BuildContext context) {
return Row(
children: <Widget>[
AnimatedOpacity(
duration: Duration(milliseconds: 500),
opacity: _opacity,
child: Container(
color: Colors.amber,
height: 100.0,
width: 100.0,
child: FlatButton(
child: Text('Tap to Fade'),
onPressed: () {
_animatedOpacity();
},
),
),
),
],
);
}
```

5. Use a duration of 500 milliseconds like in the previous animations.

```
duration: Duration(milliseconds: 500),
```

6. The animation animates the `opacity` value of the `AnimatedOpacity` child widget, which is a `Container` in this case. Depending on the value of `opacity`, the `Container` widget fades out or fades in. An `opacity` value of 1.0 is fully visible, and an `opacity` value of 0.0 is invisible. You are going to animate the `_opacity` variable from 1.0 to 0.3, and vice versa.

```
opacity: _opacity,
```

7. Add a `Container` widget as a child of the `AnimatedOpacity` widget. Add a `FlatButton` widget as a child of the `Container` widget with `onPressed` calling the `_animatedOpacity()` method.

```
child: Container(
color: Colors.amber,
height: 100.0,
width: 100.0,
child: FlatButton(
child: Text('Tap to Fade'),
onPressed: () {
_animatedOpacity();
},
),
),
```


Here's the full `_buildAnimatedOpacity()` widget class:

```
import 'package:flutter/material.dart';
class AnimatedOpacityWidget extends StatefulWidget {
const AnimatedOpacityWidget({
Key key,
}) : super(key: key);
@override
_AnimatedOpacityWidgetState createState() => _AnimatedOpacityWidgetState();
}
class _AnimatedOpacityWidgetState extends State<AnimatedOpacityWidget> {
double _opacity = 1.0;
void _animatedOpacity() {
setState(() {
_opacity = _opacity == 1.0 ? 0.3 : 1.0;
});
}
@override
Widget build(BuildContext context) {
return Row(
```

```
children: <Widget>[
AnimatedOpacity(
duration: Duration(milliseconds: 500),
opacity: _opacity,
child: Container(
color: Colors.amber,
height: 100.0,
width: 100.0,
child: FlatButton(
child: Text('Tap to Fade'),
onPressed: () {
_animatedOpacity();
},
),
),
),
],
);
}
}
```

8. Continue editing the `home.dart` file and import the `animated_opacity.dart` file to the top of the page.

```
import 'package:flutter/material.dart';
import 'package:ch7_animations/widgets/animated_container.dart';
```



```
import 'package:ch7_animations/widgets/animated_cross_fade.dart';
import 'package:ch7_animations/widgets/animated_opacity.dart';
```

9. Just after the `AnimatedCrossFadeWidget()` widget class, add a `Divider()` widget. On the next line, add the call to the widget class `AnimatedOpacityWidget()`.

```
body: SafeArea(
child: Column(
children: <Widget>[
AnimatedContainerWidget(),
Divider(),
AnimatedCrossFadeWidget(),
Divider(),
AnimatedOpacityWidget(),
],
),
),
```

USING ANIMATIONCONTROLLER

The `AnimationController` class gives you increased flexibility in animation. The animation can be played forward or reverse, and you can stop it. The `fling` animation uses a physics simulation like a spring.

The `AnimationController` class produces linear values for a giving duration, and it tries to display

a new frame at around 60 frames per second. The `AnimationController` class needs a `Ticker-Provider` class by passing the `vsync` argument in the constructor. The `vsync` prevents off-screen animations from consuming unnecessary resources. If the animation needs only one `AnimationController`, use `SingleTickerProviderStateMixin`. If the animation needs multiple `AnimationController`s, use `TickerProviderStateMixin`. The `Ticker` class is driven by the `ScheduleBinding.scheduleFrameCallback` reporting once per animation frame. It is trying to sync the animation to be as smooth as possible.

The `AnimationController` default object ranges from 0.0 to 1.0, but if you need a different range, you can use the `Animation` class (using `Tween`) to accept a different type of data. The `Animation` class is initiated by setting the `Tween` class (in-betweening) `begin` and `end` property values. For example, you have a balloon that floats from the bottom to the top of the screen, and you would set the `Tween` class `begin` value of 400.0, the bottom of the screen and the `end` value of 0.0, the top of the screen. Then you can chain the Tween `animate` method, which returns an `Animation` class. Simply put, it animates the Tween based on the animation, such as a `CurvedAnimation` class.

The `AnimationController` class at first can seem complex to use because of the different classes needed. The following are the basic steps that you take to create a custom animation (shown in Figure 7.1) or, eventually in the example, multiple animations running at the same time.

1. Add `AnimationController`.
2. Add `Animation`.
3. Initiate `AnimationController` with `Duration` (milliseconds, seconds, and so on).
4. Initiate `Animation` with `Tween` with `begin` and `end` values and chain the `animate` method with a `CurvedAnimation` (for this example).
5. Use the `AnimatedBuilder` with `Animation` using a `Container` with a balloon to start `Animation` by calling the `AnimationController.forward()` and `.reverse()` to run the animation backward. The `AnimatedBuilder` widget is used to create a widget that performs a reusable animation.

As you can see, once you break down the steps, it becomes more manageable and less complicated.

Creating the AnimationController App

In this project, you'll animate a balloon that starts small at the bottom of the screen, and as it inflates, it floats toward the top, giving some nice spring animations. By using a `GestureDetector` tap on the balloon, the animation reverses, showing the balloon deflating and floating downward to the bottom of the screen. Every time the balloon is tapped, the animation starts again.

1. Create a new Flutter project and name it `ch7_animation_controller`. You can follow the instructions in Chapter 4. For this project, you need to create only the `pages`, `widgets`, and `assets/images` folders.

FIGURE 7.1: What you're building with AnimationController

2. Open the `pubspec.yaml` file, and in `assets` add the `images` folder.

```
# To add assets to your application, add an assets section, like this:
assets:
- assets/images/
```

Add the folder `assets` and subfolder `images` at the project's root and then copy the `Beginning-GoogleFlutter-Balloon.png` file to the `images` folder.

3. Click the Save button, and depending on the editor you are using, this automatically runs `flutter packages get`. Once finished, it shows a message of `Process finished with exit code 0`. It does not automatically run the command for you, though. Open the Terminal window (located at the bottom of your editor) and type **`flutter packages get`**.
4. Create a new Dart file in the `widgets` folder. Right-click the `widgets` folder, select New ➪ Dart File, enter **`animated_balloon.dart`**, and click the OK button to save.
5. Import the `material.dart` library, add a new line, and then start typing **`st`**. The autocompletion help opens. Select the `stful` abbreviation and give it a name of `AnimatedBalloonWidget`.
6. It is appropriate to create the `AnimationController` and `Animation` variables before you can reference them in your code. Declare `TickerProviderStateMixin` to the `_AnimatedBalloonWidgetState` class by adding `with TickerProviderStateMixin`. The `AnimationController` `vsync` argument will use it. The `vsync` is referenced by `this`, meaning this reference of the `_AnimatedBalloonWidgetState` class.

```
class _AnimatedBalloonWidgetState extends State<AnimatedBalloonWidget> with
TickerProviderStateMixin {
```

7. After the class `_AnimatedBalloonWidgetState extends State<AnimatedBalloonWidget>` and before `@override`, create two `AnimationController`s to handle the duration of the balloon floating upward and the inflation of the balloon inflating. Create two `Animation`s to handle the actual movement range and inflation size. Override the `initState()` and `dispose()` methods to initiate `AnimationController` and dispose of them when the page closes.

An important note about using two `AnimationController`s is that you could have used one `AnimationController` and made use of the `Interval() curve` to stagger `Animation`. A staggered animation uses `Interval()` to `begin` and `end` animations sequentially or to overlap one another. In the "Using Staggered Animations" section of this chapter, you'll create a staggered animation app.

```
class _AnimatedBalloonWidgetState extends State<AnimatedBalloonWidget> with
TickerProviderStateMixin {
AnimationController _controllerFloatUp;
AnimationController _controllerGrowSize;
Animation<double> _animationFloatUp;
Animation<double> _animationGrowSize;
@override
void initState() {
super.initState();
```



```
_controllerFloatUp = AnimationController(duration: Duration( seconds: 4),
vsync: this);
_controllerGrowSize = AnimationController(duration: Duration(seconds: 2),
vsync: this);
}
@override
void dispose() {
_controllerFloatUp.dispose();
_controllerGrowSize.dispose();
super.dispose();
}
```

Note that in the `initState()` method you could call the `Animation` class after the `Animation-Controller` to start the animation as the page loads, but you are going to place the `Animation` class in `Widget build(BuildContext context)` instead. The reason for this is to use the `MediaQuery` class to obtain the screen size to place and size the balloon accordingly. The `initState()` method does not contain the `Widget context` object that `MediaQuery` requires. However, as `Widget build(BuildContext context)` is called, the animation starts on page load. If the user rotates the device, `Widget build(BuildContext context)` is called again, and the balloon will resize itself accordingly and run the animation if the balloon is located at the bottom of the screen.

8. After `Widget build(BuildContext context)`, create the balloon height, width, and page bottom position by using `MediaQuery.of(context).size`. I arrived at the following formulas for calculating the dimensions by testing different options to get the best look depending on different devices' screen size and orientation:

```
Widget build(BuildContext context) {
double _balloonHeight = MediaQuery.of(context).size.height / 2;
double _balloonWidth = MediaQuery.of(context).size.height / 3;
double _balloonBottomLocation = MediaQuery.of(context).size.height -
_balloonHeight;
```

Create the `Animation` class by declaring a `Tween` with a `CurvedAnimation`. For the `_animation-FloatUp` value, the `begin` property is the `_balloonBottomLocation` variable with the `end` property at 0.0. This means the floating-upward animation starts at the bottom of the screen and moves all the way to the top.

For the `_animationGrowSize` value, the `begin` property is 50.0 pixels. Set the `end` value to the `_balloonWidth` variable. This means you can start the balloon width at 50.0 pixels and increase it to the maximum `_balloonWidth` value calculated by `MediaQuery`. Use the `elasticInOut` curve to give it a beautiful spring animation that looks like a quick inflation of air.

9. Call both `_controllerFloatUp.forward()` and `_controllerGrowSize.forward()` to start the animation.

```
_animationFloatUp = Tween(begin: _balloonBottomLocation, end: 0.0).
animate(CurvedAnimation(parent: _controllerFloatUp, curve: Curves.fastOutSlowIn));
_animationGrowSize = Tween(begin: 50.0, end: _balloonWidth).
animate(CurvedAnimation(parent: _controllerGrowSize, curve: Curves.elasticInOut));
```

```
_controllerFloatUp.forward();
_controllerGrowSize.forward();
}
```

10. Create `AnimatedBuilder` and let's take a look at a high-level structural breakdown for the `AnimatedBuilder` arguments that show how `AnimatedBuilder child` (`GestureDetector` with `Image`) is passed to the `builder`, which is the widget to receive the animation.

```
return AnimatedBuilder(
animation: _animationFloatUp,
builder: (context, child) {
return Container(
child: child,
);
},
child: GestureDetector(/* Image*/) */
);
```

The `AnimatedBuilder` constructor passes the `animation` argument `_animationFloatUp`. For the `builder` argument, return a `Container` widget with the `child` property of `child`. I know that sounds weird for the `child` property, but this is the way the builder smartly redraws the `child` control being animated. The `child` widget being passed is declared next, and it will contain a `GestureDetector`
widget and a `child` widget of `Image` (balloon).
The `AnimatedBuilder` constructor has the `animation`, `builder`, and `child` arguments.
Add to the `GestureDetector()` widget the `onTap` property. For the `GestureDetector()` widget, the `onTap` property checks for `_controllerFloatUp.isCompleted` (meaning the animation is done), and if yes, it starts the animation in reverse. This will deflate the balloon and start floating it down to the bottom of the page. The `else` portion handles the balloon already at the bottom of the screen and starts the animation forward by floating upward and inflating the balloon back to normal size.

11. Add to the `GestureDetector child` property a call to the `Image.asset()` constructor that loads the `BeginningGoogleFlutter-Balloon.png` file and sets `height` to `_balloonHeight` and `width` to `_balloonWidth`.

```
return AnimatedBuilder(
animation: _animationFloatUp,
builder: (context, child) {
return Container(
child: child,
margin: EdgeInsets.only(
top: _animationFloatUp.value,
),
```



```
width: _animationGrowSize.value,
);
},
child: GestureDetector(
onTap: () {
if (_controllerFloatUp.isCompleted) {
_controllerFloatUp.reverse();
_controllerGrowSize.reverse();
}
else {
_controllerFloatUp.forward();
_controllerGrowSize.forward();
}
},
child: Image.asset(
'assets/images/BeginningGoogleFlutter-Balloon.png',
height: _balloonHeight,
width: _balloonWidth),
),
);
```

12. Open the `home.dart` file and import the `animated_balloon.dart` file to the top of the page.

```
import 'package:flutter/material.dart';
import 'package:ch7_animation_controller/widgets/animated_balloon.dart';
```

13. Add to the body a `SafeArea` with `SingleChildScrollView` as a `child`.

14. Add `Padding` as a child of the `SingleChildScrollView`.
15. Add a `Column` as a `child` of `Padding`.
16. In the `Column` children, add the call to the widget class `AnimatedBalloonWidget()`. Note I am using the `NeverScrollableScrollPhysics()` to stop the `SingleChildScrollView` to scroll content.

```
body: SafeArea(
child: SingleChildScrollView(
physics: NeverScrollableScrollPhysics(),
child: Padding(
padding: EdgeInsets.all(16.0),
child: Column(
children: <Widget>[
AnimatedBalloonWidget(),
],
),
),
),
),
```

Lab Exercises:

Exercise 1: Color & Size Toggle (Beginner)

Goal: Understand basic property animation.

Task:

- Create a square `Container` (100×100).
- On button tap:
  - Change its color (blue ↔ red)
  - Change its size (100 ↔ 200)

Constraints:

- Use `AnimatedContainer`
- Duration: 1 second
- Curve: `Curves.easeInOut`

Hint:
Change values inside `setState()`.

Exercise 2: Shape Morphing Card

Goal: Animate border radius and alignment.

Task:

- A card aligned to the **left**
- On tap:
  - Move it to the **right**
  - Change border radius from `0` to `30`

    - Add/remove shadow

Bonus:
Animate margin or padding as well.

🌓 Part B: AnimatedOpacity

Exercise 3: Fade In / Fade Out Text

Goal: Learn opacity transitions.

Task:

- Display a text: *"Welcome to Flutter Animations"*
- Button toggles opacity between `0.0` and `1.0`
- Duration: 800 ms

Restriction:
Do **not** remove the widget—only fade it.

Exercise 4: Image Reveal

Goal: Combine opacity with layout.

Task:

- Show a placeholder icon
- On button press:
    - Fade in an image
    - Fade out the placeholder

Hint:
Use **two AnimatedOpacity widgets** stacked using `Stack`.

□ Part C: AnimatedCrossFade

Exercise 5: Login ↔ Profile View

Goal: Understand cross-fading between widgets.

Task:

- Two widgets:
  - Login button
  - User profile card
- On button tap, crossfade between them

Requirements:

- Use `AnimatedCrossFade`
- Duration: 1 second
- Use `CrossFadeState.showFirst/showSecond`

Exercise 6: Expandable FAQ Tile

Goal: Crossfade between compact and expanded UI.

Task:

- First child: Question only
- Second child: Question + answer
- Tap toggles expanded view

Bonus:
Combine with `AnimatedSize` if you want extra smoothness.

⚫ Part D: AnimationController (Explicit Animations)

Exercise 7: Pulsing Circle

Goal: Learn `AnimationController`.

Task:

- Create a circle that continuously grows and shrinks
- Scale: 0.8 → 1.2
- Loop animation

Requirements:

- Use `SingleTickerProviderStateMixin`
- Use `AnimationController`
- Use `Tween<double>`

Exercise 8: Rotating Icon on Button Press

Goal: Control animation manually.

Task:

- Display a refresh icon
- On button press:
  - Rotate it **360°**
  - Stop after one full rotation

Hint:
Use `AnimatedBuilder` or `RotationTransition`.

Exercise 9: Custom Fade + Slide Animation

Goal: Combine multiple animations.

Task:

- A widget enters from bottom
- Simultaneously:
  - Fades in
  - Slides up

Requirements:

- One `AnimationController`
- Two Tweens:
  - `Offset`
  - `double` (opacity)

🧠 Additional Exercise (All-in-One)

Exercise 10: Animated Theme Switcher

Goal: Mix implicit and explicit animations.

Task:

- Toggle between Light and Dark mode
- Animate:
  - Background color (`AnimatedContainer`)
  - Icon fade (`AnimatedOpacity`)
  - Icon rotation (`AnimationController`)

Expected Outcome:
Smooth, polished UI transition like a real app setting.